ADHUSH · BEGINNER BUILD

Build the AdHush Box

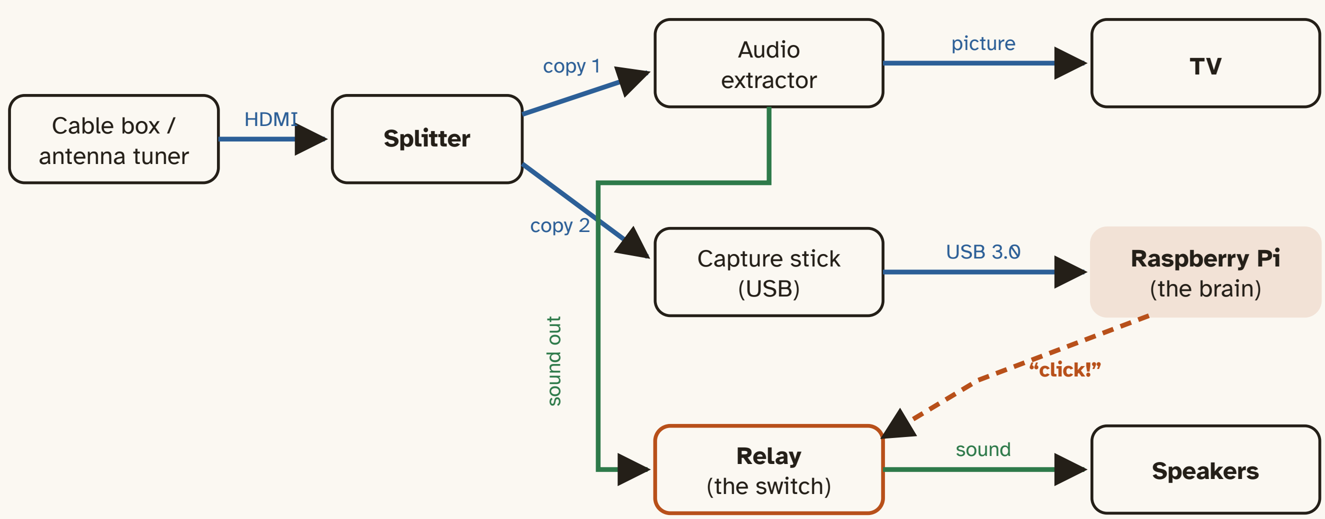
A little box that sits between your cable box and your TV, **watches for commercials, and clicks the sound off** until your show comes back. No soldering. About an afternoon. Around **$120–160** in parts. If you can build a LEGO set from instructions and type commands carefully, you can build this.

5 VOLT RELAY EDITION — the kind stores actually stock

> ⚡ **Safety rules — read these first**
>
> Everything in this project runs on **low-voltage USB power**. The biggest voltage you will ever touch is 5 volts — about the same as a phone charger cable.
>
> **Never** open, cut, or wire anything that plugs straight into a wall outlet. The only cable you cut in this whole build is a thin headphone-style audio cable, and it must be *unplugged from everything* while you cut it.
>
> Where you see the **ADULT CHECK** tag, stop and have a grown-up look at your work *before* you plug in any power.

★ The big picture

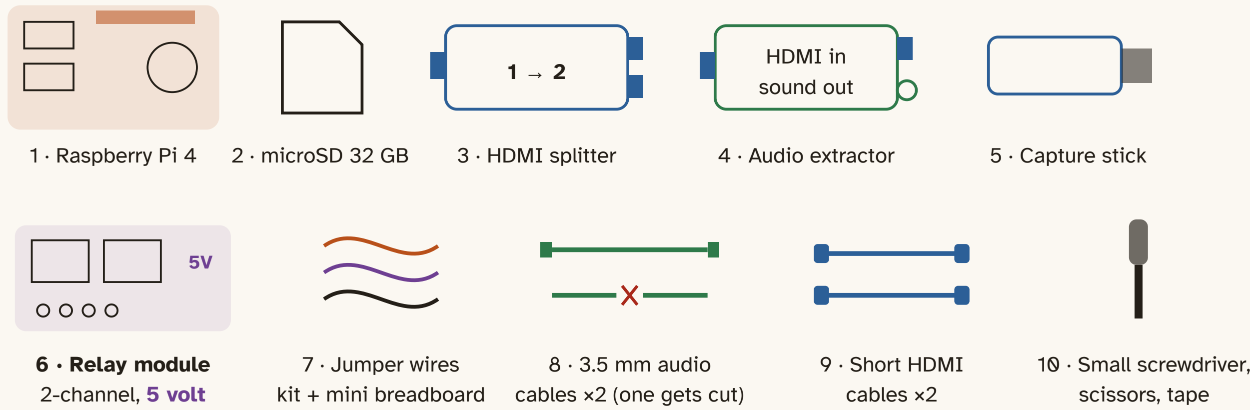
The TV signal gets **split into two copies**. One copy goes to your TV like normal. The other copy goes to the Raspberry Pi, which watches it. The **sound** takes a detour through a little switch called a **relay** — and the Pi clicks that switch whenever a commercial starts.



Blue = picture cables. Green = sound. Orange dashed = the Pi telling the relay to mute. The TV always gets its picture — only the sound gets switched.

picture (HDMI / USB) · sound · control signal · 5 V power

The parts



Lay everything out before you start, like the first page of a LEGO manual. Checking the list now is much easier than finding a missing part halfway through.

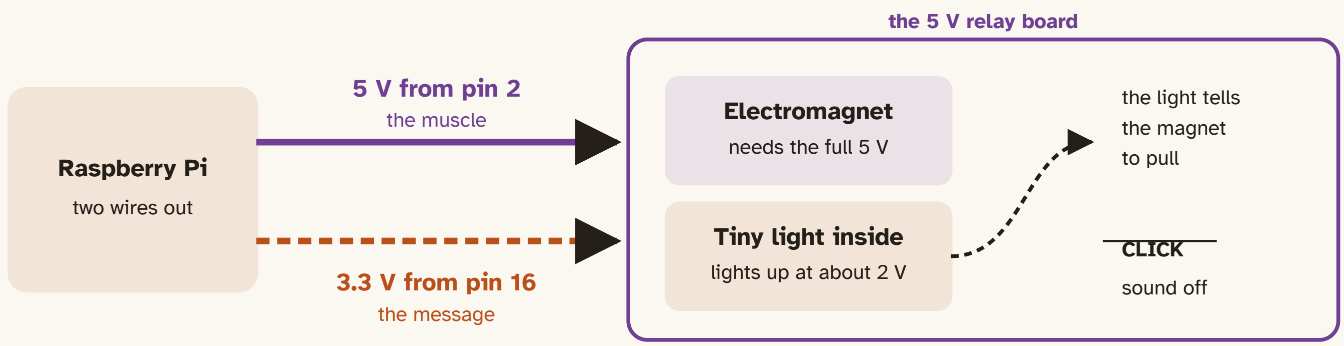
Part	What it does	≈ Price
Raspberry Pi 4 (2 GB or more) + official USB-C power supply	The brain — a tiny computer	$45–60
microSD card, 32 GB	The brain's memory	$8
Powered HDMI splitter, 1 in → 2 out	Makes a copy of the picture	$15
HDMI audio extractor (HDMI in → HDMI out + headphone/RCA out)	Pulls the sound onto a normal audio cable	$15–20
USB HDMI capture stick (must say "UVC"; USB 3.0 is best)	Lets the Pi see the copied picture	$8–25
Relay module, 2-channel, 5 V, with screw terminals — opto-isolated is best. This is the kind hobby shops keep in stock.	The click-switch that cuts the sound	$7
Jumper wire kit with a mini breadboard — you will use 2 female-to-female and 3 male-to-female wires	Connect the Pi to the relay, and send the one mute signal to *both* relay channels (Step 5 explains)	$8–10
Two 3.5 mm audio cables (one gets cut in half) — or, to skip the cutting, two "3.5 mm plug to screw terminal" adapters	Carry the sound	$6
2 short HDMI cables	Connect the boxes	$10
Small screwdriver, scissors, electrical tape	Tools you probably already have	—

Borrow for setup day: a keyboard, a mouse, a monitor or spare TV, and your Wi-Fi password. Speakers or a soundbar with an aux input sound best; your TV's own "audio in" jack works too.

Why a 5 volt relay?

Older versions of this guide asked for a **3.3 volt** relay module, because 3.3 volts is what the Pi's little signal pins put out. The trouble is that almost nobody sells 3.3 V modules in a store — walk into a hobby shop and every relay board on the shelf says **5 V**.

Good news: a 5 V board works, and this guide is written for it. Here is the trick — a relay board has **two separate jobs**, and they need different amounts of electricity.

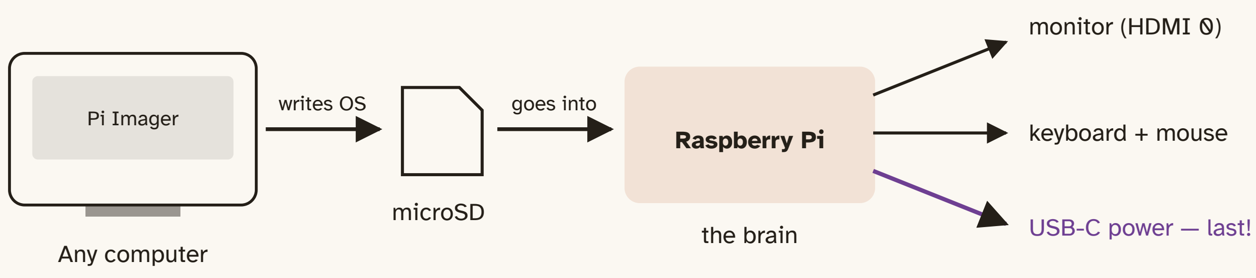


The **5 V wire** does the heavy lifting. The Pi's 3.3 V signal wire only has to light a tiny lamp inside the board, and that lamp needs about 2 volts — so 3.3 volts is plenty. That is why a 5 V board works from a 3.3 V Pi.

What changed from the old 3.3 V guide

	Old 3.3 V guide	This 5 V guide
Relay module	3.3 V logic (hard to buy)	**5 V (on every shelf)**
VCC wire	pin 1 — 3.3 V	**pin 2 or pin 4 — 5 V**
GND, signal, sound wiring	pin 6, pin 16, COM + NC	all unchanged
Settings	`active_high = true`	the same, unless the test comes out backwards

1 Wake up the Pi ADULT CHECK



Setup day only: the monitor, keyboard, and mouse all go back when you're done. Plug the power in *last* — a Pi turns on the moment it gets power.

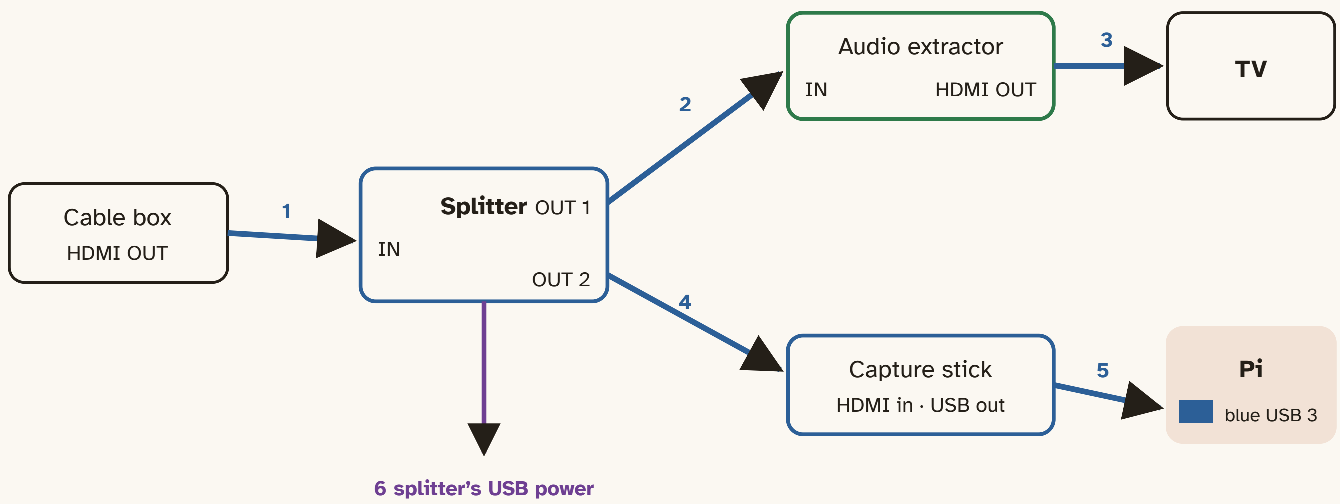
1. On a computer, install **Raspberry Pi Imager** (free, from raspberrypi.com). Put in the microSD card and write **Raspberry Pi OS** to it. In the imager's settings gear, set a username, a password, and your Wi-Fi name and password.
2. Move the card into the Pi. Connect the monitor, keyboard, and mouse. Power last. Wait for the desktop to appear.
3. Open the black **Terminal** window and type these four lines, pressing Enter after each one. Type them exactly — capital letters matter:

```
git clone https://github.com/socrtwo/adhush
cd adhush
sudo scripts/install-pi.sh
cp config/adhush-passthrough.example.toml config/adhush.toml
```

That last line makes the box's settings file. Leave it alone for now — the defaults match this build. You'll come back to it in Step 6 only if the relay turns out to click backwards.

2 Build the picture path

No tools needed — this step is all plugging in cables. You are making the copy of the TV signal that the Pi will watch.



Follow the numbers in order. Every cable in this picture is an ordinary HDMI or USB cable — nothing gets cut in this step.

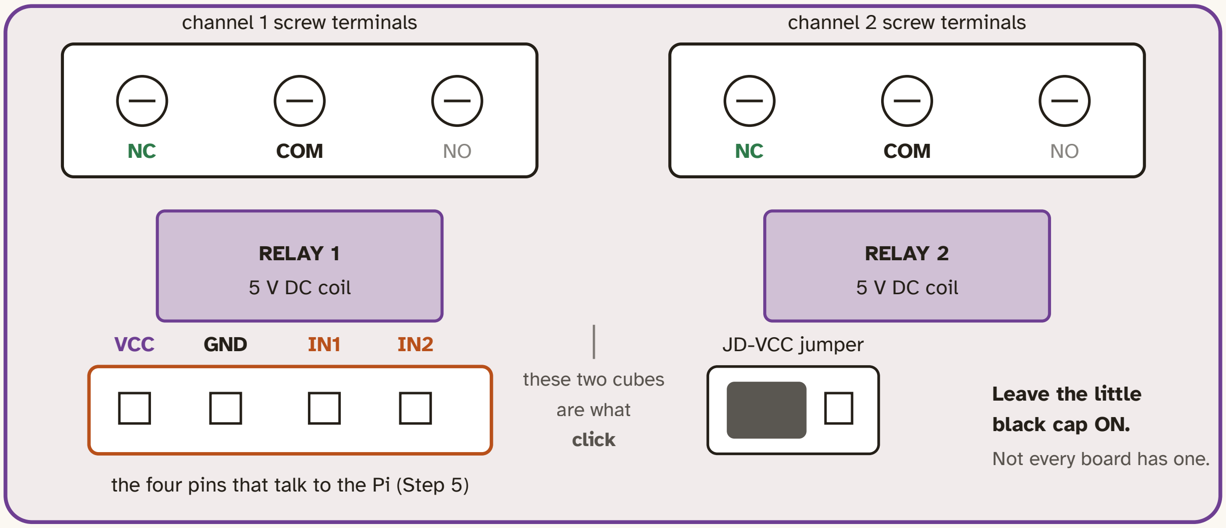
1. Unplug the HDMI cable that runs from your **cable box to your TV**, and plug that end into the **splitter's IN** instead.
2. New cable: splitter **OUT 1** → audio extractor **IN**.
3. New cable: extractor **HDMI OUT** → your **TV**. (Picture is back!)
4. New cable: splitter **OUT 2** → the **capture stick**.
5. Plug the capture stick into a **blue USB 3.0** port on the Pi.
6. Connect the splitter's little USB power cord.

Test it: turn on the TV. You should see your channels exactly like before. Black screen? Try the splitter's other output, or swap in a different HDMI cable.

3 Meet your relay board

Before any cutting, get to know the board. A 2-channel 5 V relay module has **two sides**: a *low-voltage side* where four little pins talk to the Pi, and a *switch side* with fat screw terminals where your sound wires go. The two sides never touch electrically — that is exactly why this is safe.

2-channel 5 V relay module (top view)



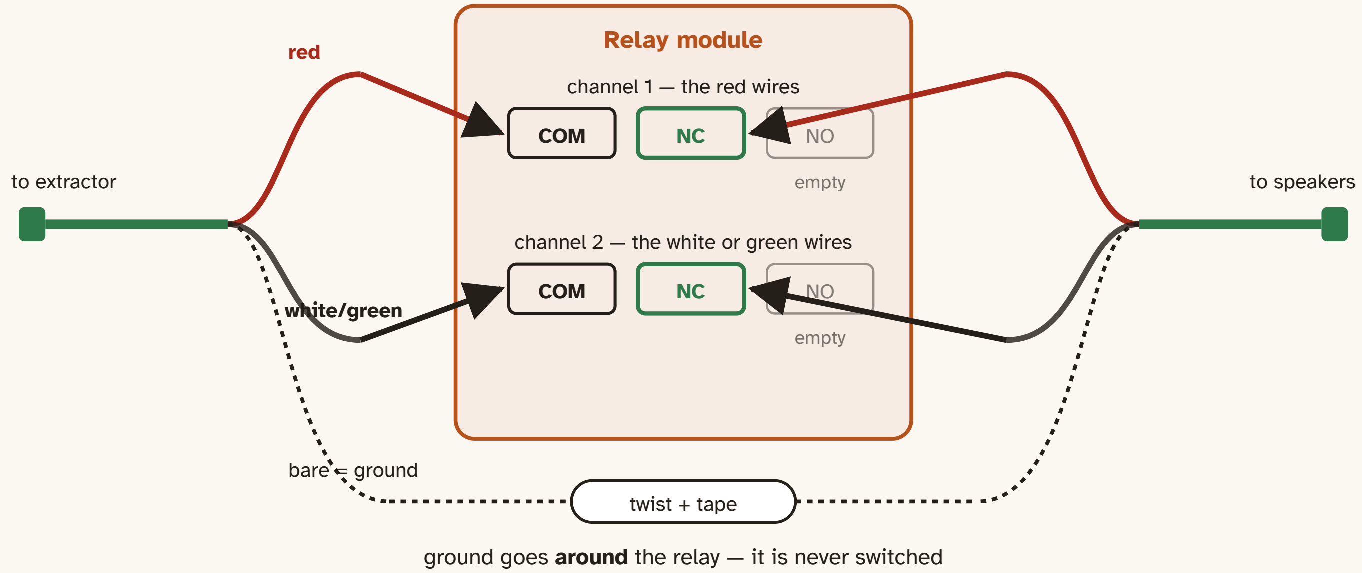
NC means “normally closed”: while the relay rests, that screw is connected to COM and sound flows through. **NO** means “normally open” — we leave those screws empty on purpose. Read the printing on your own board; the three screws are not always in this order.

The one rule that makes this safe to leave running

Sound goes through **COM + NC**, never NO. That way, if the Pi crashes, loses power, or you unplug the whole thing, the relay falls back to resting and your sound comes **back on**. A broken AdHush box should be a box that does nothing — not a box that leaves your TV silent.

4 Build the sound path ADULT CHECK

Now the scissors. Sound will travel: extractor → **through the relay** → speakers. Make sure the cable you are about to cut is unplugged from everything first.



The heart of the build. Two sound wires, two channels, four screws — and the two **NO** screws stay empty.

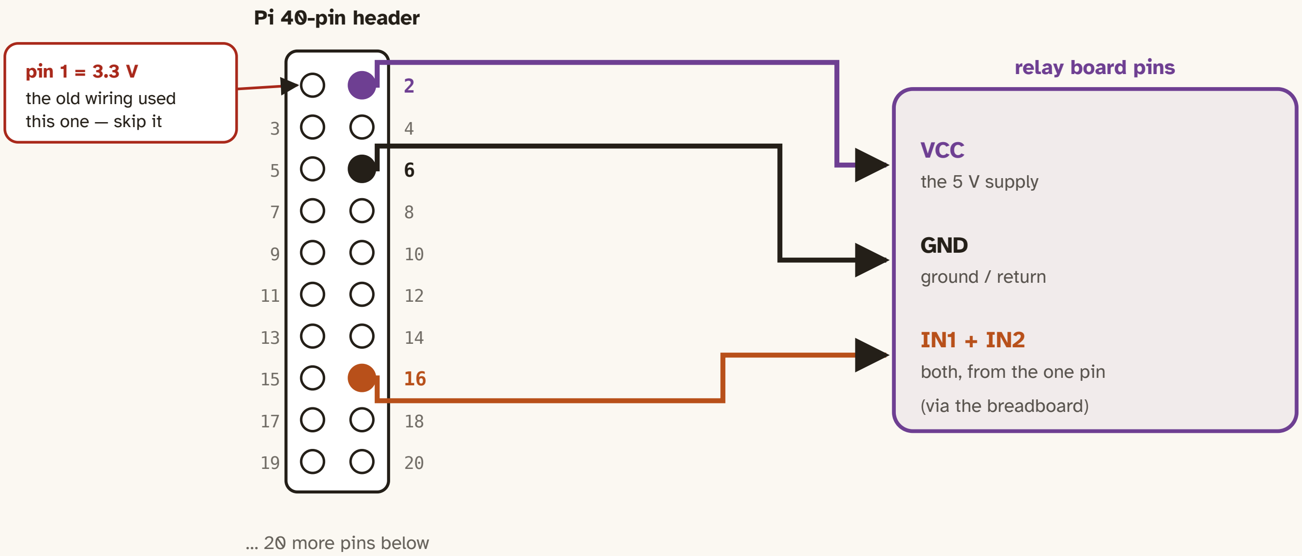
1. Cut one audio cable in the middle. Strip about 2 cm of the outer cover off each cut end. Inside are three small wires: one **red**, one **white or green**, and a bare copper one. The bare one is **ground**. (Some cables use a copper mesh instead; gather it and twist it into a strand.)
2. Look closely at the strands. If they are shiny and look painted, they are — scrape the last 1 cm with sandpaper or the back of a knife until dull copper shows. Painted strands will not make contact in a screw terminal.
3. Twist the two **ground** wires together and wrap the joint in tape. Ground is never switched.
4. Screw the **red** wires into channel 1's **COM** and **NC**. It does not matter which red goes to which of those two.
5. Screw the **white/green** wires into channel 2's **COM** and **NC**. Leave every **NO** screw empty.
6. Give each wire a gentle tug. If one pulls out, the screw isn't tight enough.
7. Plug one half of the cut cable into the **extractor's audio out** and the other half into your **speakers**.

No-scissors option

Two **"3.5 mm plug to screw terminal"** adapters (about $6 a pair) do the same job as the cut cable, with nothing to cut or scrape. Plug one adapter into the extractor and one into the speakers. Each has three screws: **L**, **R**, and **G** (ground). Run short wires L→relay channel 1→L, R→channel 2→R, and join the two G screws to each other with one wire. That is the whole sound path.

5 Connect the Pi to the relay ADULT CHECK

Five little jumper wires and a breadboard, and this is the part where the **5 volt** change matters. Power comes from the Pi's **5 V pin**. The go/stop signal comes from **GPIO 23**, which lives on **physical pin 16**.



Purple = **5 V power** from pin 2. Black = ground from pin 6. Orange = the mute signal from pin 16 (GPIO 23), which has to reach **both** IN1 and IN2.

pin 1 is the corner pin nearest the microSD card slot and **pin 2** is right beside it; odd numbers run down one side, even numbers down the other. Count twice.

Pi pin	What it is	Goes to
`pin 2`	**5 V power** (pin 4 is also 5 V — either works)	`VCC`
`pin 6`	Ground	`GND`
`pin 16`	GPIO 23 — the mute signal, 3.3 V	`IN1` ***and*** `IN2`
`pin 1`	3.3 V — what the old 3.3 V relay used	nothing

Getting one signal into two IN pins

Both channels need the *same* signal, because left and right sound must switch at the same instant. But one Pi pin only holds one jumper socket. So the signal takes a hop through the **mini breadboard**: one male-to-female wire from pin 16 into any breadboard row, then two more male-to-female wires from that *same row* out to IN1 and IN2. Every hole in a row is joined inside the board, so all three wires are connected. (A **1-channel DPDT** board, if you find one, switches both sound wires from a single IN pin and needs no breadboard.)

ADULT CHECK **before power-on**

Read this list out loud with a grown-up, pointing at each thing:

- The sound wires are on **COM** and **NC**. Both **NO** screws are empty.
- The two ground wires are twisted together and taped.
- VCC goes to **pin 2** (or pin 4) — **5 V**. Not pin 1.
- GND goes to **pin 6**. The signal goes to **pin 16**.
- No bare metal is touching any other bare metal anywhere.
- The little black **JD-VCC cap**, if your board has one, is still on.

Then power up the Pi. You should hear the TV through your speakers. Turn the TV's own volume all the way down — the relay path is the sound now.

6 Wake it up

In the Pi's terminal, inside the `adhush` folder, run these one at a time. Start with the relay test — it is the fastest way to know your wiring is right.

```
adhush probe --active   # sound should cut out for 2 seconds, then come back
adhush calibrate        # run this while a SHOW is on, not an ad
adhush run              # start watching!
```

During `adhush probe --active` you should hear one *click*, two seconds of silence, then another *click* and the sound returns. That is the whole build, proven.

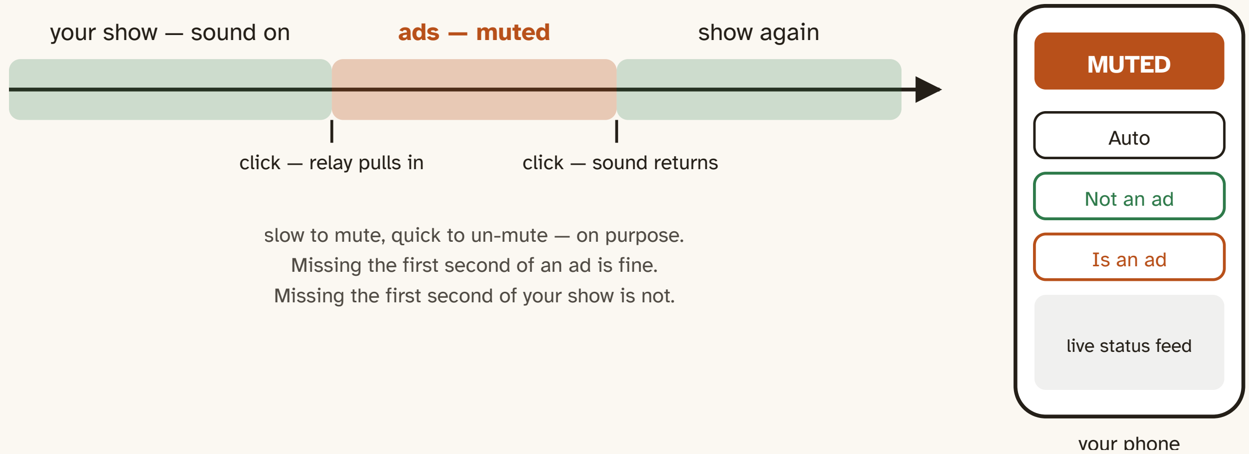
If the relay works backwards — the one 5 V gotcha

Some 5 V boards mute when the signal goes *high*. Others mute when it goes *low*. You cannot tell from the outside, so you test. If your sound is **off at rest** and comes **on** during the two-second test, your board is the other kind. One-line fix, not a wiring mistake.

Open the settings file: `nano config/adhush.toml`, find the `[control.relay_hdmi]` part, and change `true` to `false`:

```
[control.relay_hdmi]
gpio = 23
active_high = false
```

Save with **Ctrl+O**, then Enter, then **Ctrl+X** to leave. Run `adhush probe --active` again — now it should be the right way round.



What success sounds like — and your phone remote. Open `platforms/web/index.html` on any device on your Wi-Fi to watch PROGRAM / MUTED live and press the two teaching buttons.

- **“Not an ad”** — press it if the box mutes your actual show. Sound comes back instantly *and* it forgets the mistake.
- **“Is an ad”** — press it during a commercial it missed, and it memorizes that ad for next time.
- The first evening it is mostly learning, and it gets noticeably sharper on ads it has heard before. To make the box start by itself whenever it has power, run `sudo systemctl enable --now adhush`.

? If something's wrong

Problem	Try this
No picture on the TV	Reseat the HDMI cables; check the splitter's power; swap OUT 1 and OUT 2.
No sound at all	Are both wires on COM + **NC** (not NO)? Are the grounds twisted together? Is each cable half plugged into the extractor and the speakers?
Sound only on one side	Only one channel is wired, or one screw is loose. Channel 1 carries red, channel 2 carries white/green.
Muted at rest, sound during the test	Your board is the other kind. Set `active_high = false` — see the *relay works backwards* box in Step 6.
Relay never clicks at all	Run `sudo systemctl start pigpiod`, then `adhush probe --active` again. Check VCC is on **pin 2**, not pin 1 — a 5 V board will not pull in on 3.3 V.
Relay buzzes or chatters	Not enough current. Use the official Pi power supply and keep the **JD-VCC cap** on. If it still buzzes, that board wants a 3.3 V-logic module instead — not your fault.
It mutes my show!	Press **Not an ad** on the phone page — it learns from that.
`adhush run` shows an error	Read the message, then run `adhush doctor` — it lists exactly what is missing.

One honest note. A few channels and streaming boxes copy-protect their HDMI signal, so the capture stick sees nothing. AdHush never tries to break that protection. For those sources the box still works in listen-only mode, using just the sound to spot ads.

Plain-text version: `docs/build-guide-beginner.md` · wiring detail and the reasoning behind the fail-with-sound design: `docs/hardware-passthrough-box.md` · building with an Arduino, ESP32, or Pico instead: `docs/build-guide-microcontrollers.md`.

AdHush mutes your own speakers in your own home. It does not modify, copy, record, or redistribute anything you watch, and it never tries to defeat content protection.